The University of Texas at Austin
Optimization

Homework 2 Solutions

Constantine Caramanis, Sujay Sanghavi

---

1. In this problem, you will formulate an explicit linear program to model the problem below. Thus, you must state what are the decision variables, and then what is the vector corresponding to the objective function, the matrix corresponding to the constraints, etc.

   A small start-up company makes three products, creatively named Widget A, Widget B and Widget C. The company has 4 available workers it has hired, and the workers have different rates as they work on each of the three widgets. Also, because of the nature of their contracts, the workers charge a different amount depending on which widget they are working on. The time it takes for worker i to make on widget of time A, B or C, and the amount they charge depending on the widget, are summarized in the two tables given.

		worker			
		1	2	3	4
widget	A	5	7	4	10
	B	6	12	8	15
	C	13	14	9	17

Table 1: Time **in minutes** it takes Worker i to make one unit of Widget j.

		worker			
		1	2	3	4
widget	A	10	8	6	9
	B	18	20	15	17
	C	15	16	13	17

Table 2: Cost for one hour of Worker i when working on Widget j.

   Suppose each worker works for 35 hours each week. Due to the company's commitments to existing customers, the company must produce at least 100 units of Widget A, 150 units of Widget B and 100 units of Widgets C. Write a linear program that will tell the company how to assign each worker in order that the demand for each widget is met, and cost to the company is minimized.

   Solution: *The decision variables will be x_{ij}, representing how much of widget i to make using worker j. Therefore we have:*

   Objective function: $\min: \ 10x_{11} + 8x_{12} + 6x_{13} + 9x_{14} + 18x_{21} + 20x_{22} + 15x_{23} + 17x_{24} + 15x_{31} + 16x_{32} + 13x_{33} + 17x_{34}$.

For constraints, we have the total time availability of each worker:

Worker 1: $5x_{11} + 6x_{21} + 13x_{31} \leq 35 \times 60$.

Worker 2: $7x_{12} + 12x_{22} + 14x_{32} \leq 35 \times 60$.

Worker 3: $4x_{13} + 8x_{23} + 9x_{33} \leq 35 \times 60$.

Worker 4: $10x_{14} + 15x_{24} + 17x_{34} \leq 35 \times 60$.

And we also have the production commitments:

Widget A: $x_{11} + x_{12} + x_{13} + x_{14} \geq 100$.

Widget B: $x_{21} + x_{22} + x_{23} + x_{24} \geq 150$.

Widget C: $x_{31} + x_{32} + x_{33} + x_{34} \geq 100$.

Finally, we need nonnegativity constraints: $\boldsymbol{x} \geq 0$.

2. Consider the following LP:

$$\begin{aligned} \min: \quad & -x_1 - x_2 \\ \text{s.t.}: \quad & x_1 + 3x_2 \leq 3 \\ & 2x_1 + x_2 \leq 3 \\ & 4x_1 + 4x_2 \geq 1 \\ x_1, x_2 \geq 0. & \end{aligned}$$

   (i) Draw the feasible set.

   (ii) Compute explicitly all extreme points. Which is the optimal one?

   (iii) Now use the level set method to find the optimal solution. That is, draw the level sets of the objective function, and show that you find the same optimal solution as you did by enumeration in the previous part of this exercise.

   Solution: *The solution is contained in Figure 1.*

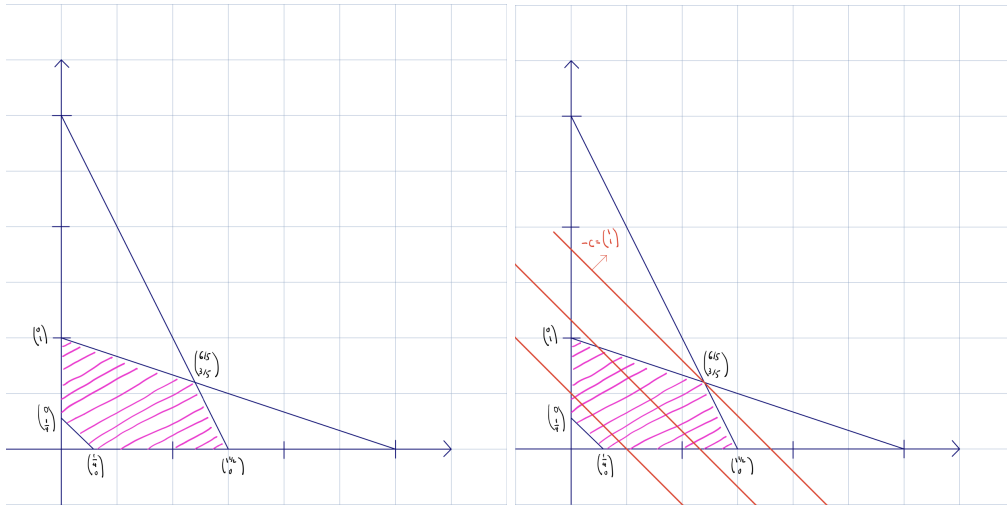
3. Formulate the following problems as LPs:

   (a) minimize $||Ax - b||_1$ subject to $||x||_\infty \leq 1$.
   Solution

$$\begin{aligned} \textit{minimize} \quad & \mathbf{1}^T y \\ \textit{subject to} \quad & -y \leq Ax - b \leq y \\ & -\mathbf{1} \leq x \leq \mathbf{1} \end{aligned}$$

   (b) minimize $||x||_1$ subject to $||Ax - b||_\infty \leq 1$.
   Solution

$$\begin{aligned} \textit{minimize} \quad & \mathbf{1}^T y \\ \textit{subject to} \quad & -y \leq x \leq y \\ & -\mathbf{1} \leq Ax - b \leq \mathbf{1} \end{aligned}$$



(a) Solution to parts (i) and (ii). (b) Solution to part (iii).

Figure 1: Solution to Problem 2

(c) minimize $||Ax - b||_1 + ||x||_\infty$
Solution

$$\begin{aligned} \textit{minimize} \quad & \mathbf{1}^T y + \alpha \\ \textit{subject to} \quad & -y \leq Ax - b \leq y \\ & -\alpha \leq x \leq \alpha \end{aligned}$$

4. Formulate the following problems as LPs.

(a) Given $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$, minimize

$$\sum_{m}^{i=1} \max\{0, a_i^T x + b_i\}$$

The variable is $x \in \mathbb{R}^n$.
Solution

$$\begin{aligned} \textit{minimize} \quad & \sum_{i=1}^{m} \gamma_i \\ \textit{subject to} \quad & 0 \leq \gamma_i \qquad i = 1, ..., m \\ & a_i^T x + b_i \leq \gamma_i \qquad i = 1, ..., m \end{aligned}$$

(b) Given $p + 1$ matrices $A_0, A_1, \ldots, A_p \in \mathbb{R}^{m \times n}$, find the vector $x \in \mathbb{R}^p$ that minimizes

$$\max_{||y||_1 = 1} ||(A_0 + x_1 A_1 + \ldots + x_p A_p) y||_1$$

Solution *First, we solve the inner max problem. For simplicity, let* $A = A_0 + x_1 A_1 + ... + x_p A_P$. *Then* $Ay = \sum_{i=1}^{n} a_i y_i$ *where* a_i *is* i^{th} *column of* A. *In consequence,*

$$||Ay||_1 = ||\sum_{i=1}^{n} a_i y_i||_1 \leq \sum_{i=1}^{n} |y_i| ||a_i||_1 \leq \max_i ||a_i||_1,$$

where the last inequality comes from the constraint $||y||_1 = 1$, *and equality is achieved by setting* $|y_i| = 1$ *where* i *is the index of the column that has the largest* l_1 *norm.*
Now we deal with a simplified problem

$$\begin{array}{ll} \textit{minimize} & \max_i ||a_i||_1 \\ \textit{subject to} & a_i = (A_0 + x_1 A_1 + ... + x_p A_P)_i \qquad i = 1, ..., m, \end{array}$$

which can be rewritten in LP with new variables associated to each column as the following:

$$\begin{array}{ll} \textit{minimize} & \gamma \\ \textit{subject to} & a_i = (A_0 + x_1 A_1 + ... + x_p A_P)_i \qquad i = 1, ..., m \\ & -y_i \leq a_i \leq y_i \qquad i = 1, ..., m \\ & \mathbf{1}^T y_i \leq \gamma \qquad i = 1, ..., m. \end{array}$$

5. We are given p matrices $A_i \in \mathbb{R}^{n \times n}$, and we would like to find a single matrix $X \in \mathbb{R}^{n \times n}$ that we can use as an approximate right-inverse for each matrix A_i , i.e., we would like to have $A_i X \approx I, i = 1, \ldots, p$. We can do this by solving the following optimization problem with X as variable:

$$\text{minimize} \max_{i=1,\ldots,p} ||I - A_i X||_\infty \tag{1}$$

Here $||H||_\infty$ is the 'infinity-norm' or 'max-row-sum norm' of a matrix H, defined as

$$||H||_\infty = \max_{i=1,\ldots,m} \sum_{j=1}^{n} |H_{ij}|,$$

if $H \in \mathbb{R}^{m \times n}$. Express problem 1 as an LP. You don't have to reduce the LP to a canonical form, as long as you are clear about what the variables are, what the meaning is of any auxiliary variables that you introduce, and why the LP is equivalent to the problem 1.
Solution *This is a two-fold optimization. Let's unfold it step by step.*

(i)

$$\begin{array}{ll} \textit{minimize} & t \\ \textit{subject to} & ||I - A_i X||_\infty \leq t \qquad i = 1, ..., p. \end{array}$$

(ii)

$$\begin{array}{ll} \textit{minimize} & t \\ \textit{subject to} & \sum_{k=1}^{n} |(I - A_i X)_{jk}| \leq t \qquad i = 1, ..., p, \;\; j = 1, ..., m \end{array}$$

where we are giving a bound to every row-sum for every i. Finally, it can be written as the following LP:

$$\begin{array}{ll} \textit{minimize} & t \\ \textit{subject to} & T_i \mathbf{1} \leq t\mathbf{1} \qquad i = 1, ..., p \\ & -T_i \leq I - A_i X \leq T_i \qquad i = 1, ..., p, \end{array}$$

where optimization variables are t, T_i, X.

6. For each of the following LPs, express the optimal value and the optimal solution in terms of the problem parameters $(c, k, d, \alpha, d_1, d_2, \ldots)$. If the optimal solution is not unique, it is sufficient to give one optimal solution.

(a)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & 0 \leq x \leq 1. \end{array}$$

The variable is $x \in \mathbb{R}^n$.
Solution *Let* $x_i = 0$ *when* $c_i \geq 0$, *and* $x_i = 1$ *when* $c_i \leq 0$. *Optimal value is* $\sum_i \min(0, c_i)$.

(b)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & -1 \leq x \leq 1. \end{array}$$

The variable is $x \in \mathbb{R}^n$.
Solution *Assign similarly as in (a). Optimal value is* $\sum_i -|c_i| = -||c||_1$.

(c)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & -1 \leq 1^T x \leq 1. \end{array}$$

The variable is $x \in \mathbb{R}^n$.
Solution *Suppose any* $x \in \mathbb{R}^n$ *that satisfies* $\mathbf{1}^T x = 0$. *Unless* $c^T x = 0$, *this problem is unbounded. Therefore, meaningful optimal value is only found when* c *is parallel to* $\mathbf{1}$. *That is,* $c = a\mathbf{1}$ *for some* $a \in \mathbb{R}$, *and* $c^T x = a\mathbf{1}^T x \geq -|a|$. *In this case, optimal value is* $-|a|$ *and optimal solution is any* x *that satisfies* $\mathbf{1}^T x = sign(a)$.

(d)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & 1^T x = 1, x \geq 0. \end{array}$$

The variable is $x \in \mathbb{R}^n$.
Solution *The best strategy is giving all weight to x_i where c_i is the minimum among all components in c. The optimal value is* $\min_{i=1,\ldots,n} c_i$.

(e)

$$\begin{array}{ll} \text{maximize} & c^T x \\ \text{subject to} & 1^T x = k, 0 \leq x \leq 1. \end{array}$$

The variable is $x \in \mathbb{R}^n$. k is an integer with $1 \leq k \leq n$.
Solution *The best strategy is giving weight to large values as much as possible. In turn, optimal value is $\sum_{i=1}^{k} c_{[i]}$ where $c_{[i]}$ denotes i^{th} largest value in c.*

(f)

$$\begin{array}{ll} \text{maximize} & c^T x \\ \text{subject to} & 1^T x \leq k, 0 \leq x \leq 1 \end{array}$$

The variable is $x \in \mathbb{R}^n$. k is an integer with $1 \leq k \leq n$.
Solution *This time, we use the same strategy to previous one, but we do not give weights to negative $c_[i]$. Optimal value is $\sum_{i=1}^{k} \max(0, c_{[i]})$.*

(g)

$$\begin{array}{ll} \text{maximize} & c^T x \\ \text{subject to} & d^T x = \alpha, 0 \leq x \leq 1. \end{array}$$

The variable is $x \in \mathbb{R}^n$. α and the components of d are positive.
Solution *We will slightly change the problem so that it becomes similar to (e). Change variables by $x_i \leftarrow x'_i/d_i$, getting*

$$\begin{array}{ll} \textit{max} & \displaystyle\sum_{i=1}^{n} (c_i/d_i) x'_i \\ \textit{subject to} & \displaystyle\sum_{i=1}^{n} x'_i = \alpha \\ & 0 \leq x'_i \leq d_i \end{array}$$

The best strategy is again to give as much weights to large values as possible. That is, sort $\frac{c_{[i]}}{d_{[i]}}$ in descending order, assign $x'_{[i]} = d_{[i]}$ from $i = 1$ to k, where k is the largest index that satisfies $\sum_{i=1}^{k} d_{[i]} \leq \alpha$. Then set $x'_{[i+1]} = \alpha - \sum_{i=1}^{k} d_{[i]}$ and 0 for remained $x_{[i]}$.

(h)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & 0 \leq x_1 \leq x_2 \leq \ldots \leq x_n \leq 1. \end{array}$$

The variable is $x \in \mathbb{R}^n$.
Solution *Optimal value is $\min_j (\sum_{i=j}^{n} c_i)$.*

(i)

$$\begin{array}{ll} \text{maximize} & c^T x \\ \text{subject to} & -y \leq x \leq y, 1^T y = k, y \leq 1. \end{array}$$

The variables are $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^n$. k is an integer with $1 \leq k \leq n$.
Solution *When the maximum is achieved, x_i should be equal to either y_i or $-y_i$, depending on the sign of c_i. Then we can rewrite the problem as*

$$\begin{array}{ll} \textit{maximize} & \sum_{i=1}^{n} |c_i y_i| \\ \textit{subject to} & \mathbf{1}^T y = k \\ & 0 \leq y \leq \mathbf{1}. \end{array}$$

Following the similar process in (e), we get the optimal value $\sum_{i=1}^{k} |c|_{[i]}$ where $|c|_{[i]}$ is the ith largest value among $|c_i|$.

(j)

$$\begin{array}{ll} \text{minimize} & 1^T u + 1^T v \\ \text{subject to} & u - v = c, u \geq 0, v \geq 0 \end{array}$$

The variables are $u \in \mathbb{R}^n$ and $v \in \mathbb{R}^n$.
Solution *The minimum possible value of $u_i + v_i$ is $|c_i|$ when $u_i - v_i = c_i$ with $u_i, v_i \geq 0$. To see that, suppose $c_i \geq 0$. Then, $u_i + v_i = c_i + 2v_i \geq c_i$ where equality is attained when $v_i = 0$. Otherwise, $u_i + v_i = -c_i + 2u_i \geq -c_i$ where similarly equality is attained when $u_i = 0$. Therefore, optimal value is $\sum_{i=1}^{n} |c_i|$.*

(k)

$$\begin{array}{ll} \text{minimize} & d_1^T u - d_2^T v \\ \text{subject to} & u - v = c, u \geq 0, v \geq 0. \end{array}$$

The variables are $u \in \mathbb{R}^n$ and $v \in \mathbb{R}^n$. We assume that $d_1 \geq d_2$.
Solution *We follow the similar procedure. Suppose $c_i \geq 0$. Then, $d_{1,i}u_i - d_{2,i}v_i = (d_{1,i} - d_{2,i})v_i + d_{1,i}c_i \geq d_{1,i}c_i$. Otherwise, $d_{1,i}u_i - d_{2,i}v_i = (d_{1,i} - d_{2,i})u_i + d_{2,i}c_i \geq d_{2,i}c_i$. In turn, we have $d_{1,i}u_i - d_{2,i}v_i \geq d_{1,i}c_i 1_{\{c_i \geq 0\}} + d_{2,i}c_i 1_{\{c_i \leq 0\}}$. Thus, the optimal solution is*

$$\sum_{i=1}^{n} (d_{1,i} 1_{\{c_i \geq 0\}} + d_{2,i} 1_{\{c_i \leq 0\}}) c_i,$$

where sgn is the sign function which gives $+1$ when $c_i \geq 0$ and -1 when $c_i \leq 0$.